

STAT 516

Central Limit Theorem

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Motivation

- ▶ Consider a binomial random variable $X \sim B(n, p)$ with $p = 0.1$ and different values of n
- ▶ It is easy to realize that a histogram of X starts rather skewed when e.g. $n = 10$ but is already more symmetric when $n = 20$
- ▶ Check additional values of $n = 50$ and $n = 100$ using the Java applet at <https://www.stat.berkeley.edu/~stark/Java/Html/BinHist.htm>
- ▶ This happens as the skewness coefficient of $X \sim B(n, p)$ is $\frac{1-2p}{\sqrt{np(1-p)}}$ that goes to zero as $n \rightarrow \infty$
- ▶ Thus, in general, $B(n, p)$ can be well approximated by $N(np, np(1 - p))$ for any fixed p when n is large
- ▶ Recall that $Bin(n, p)$ is a sum of n $Ber(p)$ random variables

Central Limit Theorem

- ▶ For $n \geq 1$ let $X_1, \dots, X_n$ be n independent random variables
- ▶ All of X_i have the same distribution with mean μ and the finite variance σ^2
- ▶ Let $S_n = X_1 + \dots + X_n$ and $\bar{X} = \frac{S_n}{n}$.
- ▶ Then, as $n \rightarrow \infty$,

1.

$$P\left(\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \leq x\right) \rightarrow \Phi(x) \forall x \in R$$

2.

$$P\left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \leq x\right) \rightarrow \Phi(x) \forall x \in R$$

- ▶ In words, for large n , $S_n \approx N(n\mu, n\sigma^2)$ and $\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$

Normal approximation to Binomial: de Moivre-Laplace Central Limit Theorem

- ▶ Let $X = X_n \sim \text{Bin}(n, p)$. Then, for any fixed p and real-valued x

$$P\left(\frac{X - np}{\sqrt{np(1-p)}} \leq x\right) \rightarrow \Phi(x)$$

as $n \rightarrow \infty$

- ▶ Thus, for any $X \sim \text{Bin}(n, p)$ we approximate $P(X \leq k)$ with $\Phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right)$

Normal approximation to Binomial: de Moivre-Laplace

Local Limit Theorem

- ▶ Sometimes, it is necessary to find the probability that a binomial random variable is exactly equal to some value
- ▶ We do this by noting that $P(X = k) = P(X \leq k) - P(X \leq k - 1)$ and stating the **local** limit theorem
- ▶ Let $X \sim \text{Bin}(n, p)$. For any fixed p and $k = 0, 1, \dots, n$

$$P(X = k) = P\left(\frac{X - np}{\sqrt{np(1-p)}} = \frac{k - np}{\sqrt{np(1-p)}}\right)$$

- ▶ The above is, approximately,

$$P(X = k) \sim \frac{1}{\sqrt{np(1-p)}} \phi\left(\frac{k - np}{\sqrt{np(1-p)}}\right)$$

Continuity Correction

- ▶ In practice, the quality of approximation improves greatly if we interpret an event $X = x$ as $x - \frac{1}{2} \leq X \leq x + \frac{1}{2}$
- ▶ This corresponds to approximating $P(X \leq k)$ as

$$P(X \leq k) \approx \Phi \left(\frac{k + \frac{1}{2} - np}{\sqrt{np(1-p)}} \right)$$

- ▶ Moreover, we also approximate

$$P(m \leq X \leq k) \approx \Phi \left(\frac{k + \frac{1}{2} - np}{\sqrt{np(1-p)}} \right) - \Phi \left(\frac{m - \frac{1}{2} - np}{\sqrt{np(1-p)}} \right)$$

Example: coin tossing

- ▶ Suppose a fair coin is tossed 100 times. What is the probability that we obtain between 45 and 55 heads?
- ▶ The number of heads is $X \sim \text{Bin}(n, p)$ with $n = 100$ and $p = 0.5$
- ▶ Thus,

$$\begin{aligned} P(45 \leq X \leq 55) &\approx \Phi\left(\frac{55.5 - 50}{\sqrt{25}}\right) - \Phi\left(\frac{44.5 - 50}{\sqrt{25}}\right) \\ &= \Phi(1.56) - \Phi(-1.56) = 0.8812 \end{aligned}$$

Example: coin tossing

- ▶ How many times do we need to toss a fair coin to be 99% sure that the percentage of heads is between 45% and 55%?
- ▶ In other words, when is the number of heads between $.45n$ and $.55n$?
- ▶ Thus,

$$.99 = \Phi\left(\frac{.55n + 0.5 - .5n}{\sqrt{.25n}}\right) - \Phi\left(\frac{.45n - 0.5 - .5n}{\sqrt{.25n}}\right)$$

- ▶ This is equivalent to

$$.99 = 2\Phi\left(\frac{.55n + 0.5 - .5n}{\sqrt{.25n}}\right) - 1$$

since $\Phi(x) - \Phi(-x) = 2\Phi(x) - 1$

Example: coin tossing



$$\Phi\left(\frac{.05n + 0.5}{\sqrt{.25n}}\right) = 0.995$$

- ▶ Since $\Phi(2.575) = 0.995$, we end up with a quadratic equation in $\sqrt{n}$:

$$0.05n - 1.2875\sqrt{n} + 0.5 = 0$$

- ▶ The needed solution is $n = 661.04 \approx 662$

Public polling: Predicting the correct winner

- ▶ In an election with two candidates, A and B , it is known that among *all* voters, 52% support A and 48% support B
- ▶ A poll of 1400 voters is done; what is the probability that the poll will predict the correct winner?
- ▶ The number of respondents in the poll who favor A is $X \sim \text{Bin}(1400, 0.52)$
- ▶ Approximately,

$$\begin{aligned} P(X > 700) &= 1 - P(X \leq 700) \approx 1 - \Phi\left(\frac{700.5 - 1400 * .52}{\sqrt{1400 \times .52 \times .48}}\right) \\ &= 1 - \Phi(-1.5) = \Phi(1.5) = .9332 \end{aligned}$$

Public Polling: Predicting the Vote Share

- ▶ Now, suppose that the proportion of those who support A among all voters is p
- ▶ A poll of n respondents is conducted
- ▶ How large n should be if with 95% probability we want to predict the true p within an error of at most 2%
- ▶ Let X be the number of polled respondents who favor A . We want to ensure that

$$P\left(\left|\frac{X}{n} - p\right| \leq 0.02\right) \geq .95$$

Public Polling: Predicting the Vote Share

- ▶ Equivalently,

$$P\left(-\frac{.02\sqrt{n}}{\sqrt{p(1-p)}} \leq \frac{X - np}{\sqrt{np(1-p)}} \leq \frac{.02\sqrt{n}}{\sqrt{p(1-p)}}\right) \geq .95$$

- ▶ By the normal approximation, the left-hand side above is

$$\begin{aligned} &\approx \Phi\left(\frac{.02\sqrt{n}}{\sqrt{p(1-p)}}\right) - \Phi\left(-\frac{.02\sqrt{n}}{\sqrt{p(1-p)}}\right) \\ &= 2\Phi\left(\frac{.02\sqrt{n}}{\sqrt{p(1-p)}}\right) - 1 \end{aligned}$$

- ▶ Therefore,

$$\frac{.02\sqrt{n}}{\sqrt{p(1-p)}} = 1.96$$

and $n = 9604p(1-p)$

- ▶ Usually, the most conservative value of $p = 0.5$ is used giving rise to $n = 2401$

General CLT example I

- ▶ Suppose a fair die is rolled n times; let X_i , $1 \leq i \leq n$ be the individual rolls
- ▶ Let $S_n = \sum_{i=1}^n X_i$; recall that, for each X_i the mean $\mu = 3.5$ and $\sigma^2 = 2.92$
- ▶ Therefore,

$$S_n \approx N(3.5n, 2.92n)$$

General CLT example I

- For $n = 100$ the probability

$$\begin{aligned} P(S_n \geq 300) &= 1 - P(S_n \leq 299) = 1 - \Phi\left(\frac{299.5 - 3.5 * 100}{\sqrt{2.92 * 100}}\right) \\ &= 1 - \Phi(-2.96) = \Phi(2.96) = .9985 \end{aligned}$$

General CLT example II

- ▶ Let n positive numbers be rounded up to their nearest integers
- ▶ The rounding error $e_i \sim U[-0.5, 0.5]$
- ▶ E.g. a tax agency rounds off the exact refund amount to the nearest integer
- ▶ Then, the total error is the agency's loss or profit due to the rounding process

General CLT example II

- ▶ Recall that $E e_i = 0$ and $V(e_i) = \frac{1}{12}$
- ▶ By the CLT, the total error

$$S_n = \sum_{i=1}^n e_i \sim N\left(0, \frac{n}{12}\right)$$

- ▶ E.g. when $n = 1000$

$$\begin{aligned} P(|S_n| \leq 20) &= P(S_n \leq 20) - P(S_n \leq -20) \\ &= P\left(\frac{S_n}{\sqrt{\frac{n}{12}}} \leq \frac{20}{\sqrt{\frac{n}{12}}}\right) - P\left(\frac{S_n}{\sqrt{\frac{n}{12}}} \leq \frac{-20}{\sqrt{\frac{n}{12}}}\right) \\ &\approx \Phi(2.19) - \Phi(-2.19) = .9714 \end{aligned}$$

- ▶ Due to cancellations of positive and negative errors the tax agency is unlikely to lose or gain much money from rounding